

Day 15 Observation Guide: Area of a Trapezoid

OBSERVER GUIDE

Grade 6 Mathematics · Professor Xavier Honablué, M.Ed. · For visiting guests and observers

Purpose of This Guide

Day 15 closes the plane-figure arc of Unit 1. Students meet the trapezoid — a quadrilateral with exactly one pair of parallel sides — and find its area two ways: by **decomposing** it into a rectangle and two triangles (Days 5 and 13), and by **composing** two copies into a parallelogram with base $b_1 + b_2$ (Day 8). Both roads arrive at $A = \frac{1}{2}(b_1 + b_2)h$, and students are expected to be able to travel either one.

This guide explains the design, maps today to the i-Ready lesson, and links every material and research source used to build it.

The Design Decision: Two Methods, Deliberately

We do not pick a favorite method today. Decomposition is the method that always works and needs no new formula — it is the fallback when a student forgets. Composition (two copies making a parallelogram) is the method that explains the formula, including why the two bases get added and why the result is halved. Students who hold both can check themselves; students who hold only the formula cannot.

Worked Examples 1 and 2 in the packet are the same trapezoid solved both ways on purpose, so the agreement of the two answers is visible rather than asserted.

Subscripts Get Taught, Not Assumed

This is the first lesson where students meet subscript notation: b_1 and b_2 . We teach it explicitly — the small numbers are name tags, not exponents and not multiplication — because an unexplained symbol is a reliable source of later confusion. Observers will hear the class read the formula aloud as "b-sub-one plus b-sub-two, times the height, cut in half."

i-Ready Classroom Mathematics Alignment

Today's lesson is drawn from i-Ready Classroom Mathematics, Grade 6, Unit 1, Lesson 3 ("Find the Area of Polygons"), taught after Lesson 4's nets so that the triangle formula from Day 13 was available for the triangular-prism net on Day 14.

i-Ready content (Grade 6, Unit 1)	Where it appears today	Standard(s)
Lesson 3, Session 1 · Explore — decomposing polygons into known shapes	Learn section 2; Worked Example 2	6.G.A.1, MP.7
Lesson 3, Session 2 · Develop — the trapezoid area formula	Learn section 3; Worked Example 1	6.G.A.1, MP.8
Lesson 3, Session 2 · Try It — missing base or height	Learn section 4; Worked Example 3; practice items 5–6	6.G.A.1, 6.EE.B.7
Lesson 3, Session 3 · Refine — composite polygons and real-world cost problems	Worked Example 4; practice items 9 and 12	6.G.A.1, 6.RP.A.3
Lesson 3 · Math Journal — compare the two methods	Closing journal prompt	MP.3

Teacher note: session numbers follow the i-Ready Classroom Mathematics Grade 6 Teacher Edition. Confirm exact page numbers against your edition before distributing this guide to observers.

Lesson Flow — 55 Minutes

Min	Segment	What the class is doing	Look for
0-5	Do Now	One triangle and one surface-area problem from Days 13-14.	Retention across two different figure types.
5-18	Explore	Cut a paper trapezoid into a rectangle + two triangles; then build the doubled parallelogram with a partner's copy.	Students stating $b_1 + b_2$ before the formula is written.
18-30	Develop	Worked Examples 1-2 — the same trapezoid, both methods, side by side.	Recognition that the two answers must agree.
30-45	Practice	Lesson & Practice packet items 1-12, including the grid trapezoid and the distractor slant.	Correct identification of the two parallel bases.
45-55	Refine & close	Error analysis (item 8: the student who forgot to halve) and the journal comparison prompt.	Students defending a preferred method with a reason.

Predictable Student Errors — and What We Do About Them

Error	Why it happens	Our move today
Forgetting to halve	The doubled parallelogram is the last picture students saw.	Ask: "is your answer the trapezoid or both copies?"
Adding a slanted side into $b_1 + b_2$	All four sides look similar in a drawing.	Highlight the two parallel sides in one color before computing.
Using a slanted side as the height	Carried over from parallelograms.	Same routine: dotted height, right-angle box, every time.
Missing base: subtracting before dividing	Steps are done out of order.	Write $2A = (b_1 + b_2)h$ as its own line; find the sum first, then subtract.

What an Observer Can Look For

- Both methods present in the room — and students saying which one they trust and why.
- The two parallel bases marked before any arithmetic.
- Subscript language used correctly out loud ("b-sub-one").
- Grid work (item 9) using coordinates to read off the bases and height.
- Real-world items finishing with a cost, not just an area.

Today's Materials — Direct Links

Every item below is also one click away on the launcher: cpamathlauncher.streamlit.app.

Material	Link
Lesson & Practice packet (12 problems + full solutions)	Day15_Lesson_Packet.pdf
Anchor chart — Area of a Parallelogram (the composition method)	Anchor_Chart_Area_of_a_Parallelogram.pdf
Day 13 Lesson & Practice packet (the decomposition method)	Day13_Lesson_Packet.pdf
Homework — IXL, Grade 6 skill "Area of trapezoids" (90% target)	ixl.com/math/grade-6/area-of-a-trapezoid

Standards Addressed

Code	Standard	Where it shows up today
6.G.A.1	Find the area of special quadrilaterals and polygons by composing into rectangles or decomposing into triangles and other shapes.	Both solution methods; composite deck problem.
6.G.A.3	Draw polygons in the coordinate plane; use coordinates to find side lengths.	Grid trapezoid (item 9).
6.EE.B.7	Solve $px = q$ problems.	Missing base and missing height (items 5–6).
MP.3	Construct viable arguments and critique the reasoning of others.	Error analysis item 8; journal comparison of methods.
MP.8	Express regularity in repeated reasoning.	Generalizing from two copies to the formula.

Research & Further Reading

Every hyperlink below opens the source directly. These are the sources used to build today's lesson.

Source	Why it is here
Common Core State Standards — Grade 6, Geometry (6.G.A.1–4)	The exact wording of the standards this unit teaches, including the "compose and decompose" language our lessons are built on.
IES / What Works Clearinghouse — Improving Mathematical Problem Solving in Grades 4–8 (full PDF)	The federal practice guide behind our daily routine: worked examples with steps, student error analysis, and multiple solution paths.
Curriculum Associates — i-Ready Classroom Mathematics	The publisher page for the curriculum this unit follows (Try–Discuss–Connect instructional routine).
i-Ready Family Center	Free family-facing explanations of each lesson — useful to hand to parents who ask how to help at home.
YouCubed at Stanford — open math tasks	Stanford Graduate School of Education tasks used for our visual and multi-strategy work.
Illustrative Mathematics — Grade 6, Unit 1, Lesson 11: Polygons	Open-licensed lesson on decomposing polygons into triangles and rectangles.
GeoGebra — Area of a Parallelogram (Discovery)	Used again today: the doubled trapezoid is a parallelogram, so the same slider argument applies.
Khan Academy — Area of a trapezoid	Short video for absent students and home review.

Unit 1's plane figures are complete after today: rectangles, parallelograms, triangles, and trapezoids, plus nets and surface area. The next assessment window draws from all of them — see the Summative Test Sessions button on the launcher.